

$$* V_r = (v - u) R \sin(\theta)$$

{ A
B

* Use H α lines to probe these velocities.

• Dark matter doesn't radiate

→ Gives us a way to "probe" the amount:

$$\text{We look at } \left(\frac{\text{Mass}}{\text{Light}} \right) \sim \left(\frac{M}{L} \right)$$

• If we look at M31 (Andromeda): we can look at the L output.

* If you look at the Milky way: large $\left(\frac{M}{L} \right)$

→ baryonic matter "radiating matter" $\sim 7\%$

→ Dark matter: $\sim 93\%$

* Also: when we look at v near the center of the MW: eccentric orbits for stars.

→ Moving faster than expected

→ oddly

→ A Supermassive Black hole:

~ 0 to $1-2$ kpc: in general assume 2 kpc

• As you go out ~ 3 kpc (Disk "diameters")

• Sun: 8 kpc

* IMF: varies so the $\rho(R)$ also varies:

$$(1-2 \text{ kpc}) \sim M(R) \sim \sqrt{R}$$

$$v^2 = \frac{G \cdot M}{R} \sim v \sim \frac{1}{\sqrt{R}}$$

• $\sim 1-3$

$M(R) \sim R$

$V \sim \text{const.}$

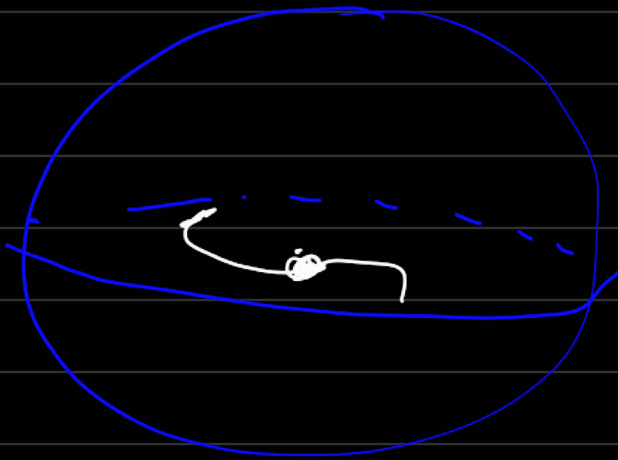
→ Here is where we can assume circular.

$$V^2 = \frac{GM(R)}{R}$$

$M(R) \sim \text{constant.}$

• Part 3: V rises again

• Part "ends of the spiral arms"
 $\sim 20 \text{ kpc}$: Dark matter dominates:



* look at other galaxies to understand if we're the weird ones.

Define the Surface-Mass-density: $\Sigma_s(R)$

$\rho(R)$: Mass density $\sim \frac{M}{V}$

$\Sigma_s(R)$: SMD $\sim \frac{M}{A}$



• Not interested in h

$$V = Ah$$
$$\Sigma_s(R) \sim \rho h = \frac{Mh}{V} = \frac{MK}{AK}$$
$$= \frac{M}{A}$$

$$*\left(\frac{M}{L}\right) \approx 1 \text{ ("typical")}$$

$$\left(\frac{M}{L}\right) \gg 1 \quad \approx 20 \text{ i. up}$$

$$\text{assume: } \frac{M}{L} \sim 50 \Rightarrow M \sim 50L$$

* WIMPS: Dark matter

HII: Star forming regions:

→ Recomb. $\Rightarrow$ H α emission

$$\text{LSR: older stars } \begin{cases} V_{\text{rot},0} \sim 20-30 \text{ m/s} \\ \text{newer: } V_{\text{rot},1} \sim 10 \text{ m/s} \end{cases}$$

$$\text{average} \sim \frac{V_0 + V_1}{2} \sim 15 \text{ m/s}$$

